

Q1) What is Machine Learning? Compare Supervised Learning with Unsupervised Learning. (5 Marks)

What is Machine Learning?

Definition:

Machine Learning (ML) is a branch of Artificial Intelligence (AI) that enables computers to learn from data and make predictions or decisions without being explicitly programmed.

Definition by Arthur Samuel:

"Machine Learning is the field of study that gives computers the ability to learn without being explicitly programmed."

Examples

- Spam email detection
- Face recognition
- Product recommendation
- Weather prediction

Difference between Supervised and Unsupervised Learning

Supervised Learning	Unsupervised Learning
Uses labeled data	Uses unlabeled data
Correct output is already known	Correct output is not known
Used for prediction	Used for finding patterns or groups
Higher accuracy	Lower accuracy
Performs classification and regression	Performs clustering and association
Example: Spam detection	Example: Customer segmentation

Examples

- **Supervised:** House price prediction, Email spam detection
- **Unsupervised:** Customer grouping, Market basket analysis

Conclusion

Supervised learning predicts known outputs using labeled data, while unsupervised learning discovers hidden patterns from unlabeled data.

Q2) Explain Applications of Machine Learning in the Real World / Data Science. (5 Marks)

Definition

Machine Learning is used in many real-life applications to automate tasks, make predictions, and improve decision-making.

Applications

1. Email Spam Detection

- Identifies spam and genuine emails.
- Example: Gmail Spam Filter.

2. Recommendation Systems

- Recommends products, movies, or songs based on user interests.
- Example: Amazon, Netflix, YouTube.

3. Healthcare

- Predicts diseases and helps doctors in diagnosis.
- Example: Cancer detection.

4. Face Recognition

- Identifies people using facial features.
- Example: Mobile face unlock.

5. Banking and Finance

- Detects fraudulent transactions.
- Predicts loan approval and credit risk.

6. Self-Driving Cars

- Detects roads, vehicles, and traffic signs.
- Helps in automatic driving.

7. Weather Forecasting

- Predicts rainfall and temperature.

8. Data Science

- Predicts future trends.
- Performs customer segmentation.
- Detects anomalies.
- Builds predictive models.

Advantages

- Saves time
- Improves accuracy
- Automates decision-making
- Handles large amounts of data

Q3) How does an ML Model Work? Explain the Steps in Developing a Machine Learning Application. (5 Marks)

Definition

A Machine Learning model learns patterns from data during training and uses the learned knowledge to make predictions on new data.

Steps in Developing an ML Application

1. Data Collection

- Collect data from databases, websites, sensors, or files.
- Good quality data improves model performance.

↓

2. Data Preprocessing

- Clean the data by removing missing values and duplicates.
- Convert data into a suitable format.

↓

3. Feature Selection/Engineering

- Select important features that affect prediction.
- Remove unnecessary features.

↓

4. Model Selection

- Choose a suitable ML algorithm.
- Example: Decision Tree, KNN, Naive Bayes, Linear Regression.

↓

5. Model Training

- Train the model using training data.
- The model learns patterns from the data.

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6. Model Testing/Evaluation

- Test the model using test data.
- Measure accuracy and performance.

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7. Prediction (Deployment)

- Use the trained model to make predictions on new data.
- Deploy the model for real-world use.

Flow Diagram

Data Collection

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Data Preprocessing

↓

Feature Selection

↓

Model Selection

↓

Model Training

↓

Model Testing

↓

Prediction / Deployment

Advantages

- Produces accurate predictions.
- Automates decision-making.
- Improves performance with more data.

Q2) Compare Machine Learning with Traditional Programming. Discuss Types of Machine Learning with Suitable Examples. (5 Marks)

A) Comparison of Machine Learning and Traditional Programming

Traditional Programming

Programmer writes all the rules.

Input + Program → Output

Works on fixed instructions.

Best for simple and well-defined problems.

Does not improve with experience.

Example: Calculator, Payroll System

Machine Learning

Machine learns rules from data.

Input + Output (Training Data) → ML Model

Learns and improves automatically.

Best for complex prediction problems.

Improves as more data is provided.

Example: Spam Detection, Face Recognition

Diagram

Traditional Programming

Input + Program

↓

Output

Machine Learning

Input + Output

↓

Training

↓

ML Model

↓

Prediction

B) Types of Machine Learning

Machine Learning is mainly divided into **four types**.

1. Supervised Learning

- Uses **labeled data**.

- The correct output is already known.
- Used for **classification** and **regression**.

Examples:

- Email spam detection
 - House price prediction
-

2. Unsupervised Learning

- Uses **unlabeled data**.
- Finds hidden patterns or groups in data.

Examples:

- Customer segmentation
 - Market basket analysis
-

3. Semi-Supervised Learning

- Uses **both labeled and unlabeled data**.
- Improves accuracy while reducing labeling cost.

Examples:

- Medical image classification
 - Face recognition
-

4. Reinforcement Learning

- An **agent** learns by interacting with the environment.
- Receives **rewards** for correct actions and **penalties** for wrong actions.

Examples:

- Self-driving cars
- Robot navigation
- Game playing (Chess)

Q3) Compare Machine Learning (ML) and Artificial Intelligence (AI). (5 Marks)

Definition of Artificial Intelligence (AI)

Artificial Intelligence (AI) is a branch of computer science that enables machines to perform tasks that normally require human intelligence, such as learning, reasoning, problem-solving, and decision-making.

Definition of Machine Learning (ML)

Machine Learning (ML) is a subset of AI that enables computers to learn from data and improve their performance without being explicitly programmed.

Difference between AI and ML

Artificial Intelligence (AI)

AI is a broad field that makes machines intelligent.

AI aims to mimic human intelligence.

AI may use rules, logic, and learning techniques.

AI can work without learning from data.

AI solves a wide variety of intelligent tasks.

Examples: Virtual assistants, Robots

Machine Learning (ML)

ML is a subset of AI that helps machines learn from data.

ML aims to learn patterns from data and make predictions.

ML mainly uses algorithms and training data.

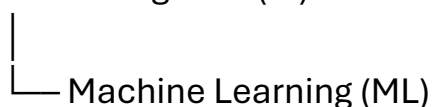
ML requires data for training.

ML mainly solves prediction and classification problems.

Examples: Spam detection, Recommendation systems

Relationship between AI and ML

Artificial Intelligence (AI)



ML is a subset of AI. Every Machine Learning system is a part of AI, but not every AI system uses Machine Learning.

Examples

Artificial Intelligence

- Virtual assistants (Siri, Google Assistant)
- Self-driving cars
- Chess-playing systems

Machine Learning

- Email spam detection
- Movie recommendation system
- House price prediction

Conclusion

Artificial Intelligence is the broader field that focuses on creating intelligent systems, while Machine Learning is a technique within AI that enables systems to learn from data and improve their performance automatically.

Q4(A) Explain Supervised, Unsupervised and Semi-Supervised Learning. (7 Marks)

1. Supervised Learning

Definition

Supervised Learning is a type of Machine Learning in which the model is trained using **labeled data**, i.e., the input data already has the correct output.

Working

1. Collect labeled data.
2. Train the model using input and output data.
3. Model learns the relationship.
4. Predicts output for new data.

Applications

- Email spam detection
- House price prediction
- Student result prediction

Advantages

- High prediction accuracy.
- Easy to evaluate performance.

Disadvantages

- Requires a large amount of labeled data.
 - Data labeling is time-consuming and costly.
-

2. Unsupervised Learning

Definition

Unsupervised Learning is a type of Machine Learning in which the model is trained using **unlabeled data**. The model finds hidden patterns or groups in the data.

Working

1. Collect unlabeled data.
2. Model analyzes similarities.
3. Forms clusters or discovers patterns.

Applications

- Customer segmentation
- Market basket analysis
- News article grouping

Advantages

- No labeled data is required.
- Finds hidden patterns in data.

Disadvantages

- Lower accuracy than supervised learning.
 - Results are difficult to evaluate.
-

3. Semi-Supervised Learning

Definition

Semi-Supervised Learning uses **both labeled and unlabeled data** for training. A small amount of labeled data guides the learning process, while a large amount of unlabeled data improves performance.

Working

1. Use a small labeled dataset.
2. Add a large unlabeled dataset.
3. Train the model using both datasets.

Applications

- Face recognition
- Medical image analysis
- Speech recognition

Advantages

- Better accuracy than unsupervised learning.
- Reduces the cost of labeling data.

Disadvantages

- More complex than supervised learning.
- Performance depends on the quality of labeled data.

Comparison Table

Supervised Learning	Unsupervised Learning	Semi-Supervised Learning
Uses labeled data	Uses unlabeled data	Uses both labeled and unlabeled data
Correct output is known	Correct output is unknown	Only some outputs are known
Used for prediction	Used for finding patterns	Used to improve prediction accuracy
Higher accuracy	Lower accuracy	Moderate to high accuracy
Example: Spam detection	Example: Customer segmentation	Example: Face recognition

Q4(B) What are the Differences Between Supervised Learning and Reinforcement Learning? (5 Marks)

Supervised Learning

Uses labeled data for training.
Correct output is already known.
Learns from input-output examples.

Reinforcement Learning

Learns by interacting with the environment.
Correct output is not provided directly.
Learns through rewards and penalties.

Supervised Learning

Objective is to predict correct output.

Feedback is immediate and direct.

Used for classification and regression.

Examples: Spam detection, House price prediction

Reinforcement Learning

Objective is to maximize total reward.

Feedback is delayed and based on rewards.

Used for sequential decision-making.

Examples: Self-driving cars, Robot navigation, Game playing

Reinforcement Learning (Short Note)

Definition

Reinforcement Learning is a type of Machine Learning in which an **agent learns by interacting with the environment**. It receives **rewards** for correct actions and **penalties** for wrong actions.

Components

- **Agent:** Learner or decision-maker.
- **Environment:** The world in which the agent operates.
- **State:** Current situation of the agent.
- **Action:** Decision taken by the agent.
- **Reward:** Feedback received after an action.

Examples

- Self-driving cars
- Robot navigation
- Chess and other game-playing AI

Q5) Explain Models of Machine Learning in Detail (General Overview). (5 Marks)

Definition

Machine Learning models are mathematical methods or algorithms that learn patterns from data and make predictions or decisions.

There are different types of Machine Learning models based on how they represent and process data.

1. Geometric Model

- Represents data as **points in geometric space**.
- Uses **distance, lines, or planes** to separate data.
- Mainly used for **classification**.

Example:

- Support Vector Machine (SVM)
- K-Nearest Neighbors (KNN)

2. Probabilistic Model

- Uses **probability** to make predictions.
- Calculates the probability of each possible outcome.
- Useful when data contains uncertainty.

Example:

- Naive Bayes Classifier
-

3. Logical Model

- Makes decisions using **logical IF-THEN rules**.
- Easy to understand and explain.
- Used for classification and decision-making.

Example:

- Decision Tree
-

4. Grouping and Grading Model

Grouping Model

- Groups similar data into clusters.
- Used in customer segmentation.

Example:

- K-Means Clustering

Grading Model

- Assigns scores or ranks to data.
- Used for prediction and ranking.

Example:

- Credit Score Prediction
 - Student Grade Prediction
-

5. Parametric and Non-Parametric Models

Parametric Model

- Has a **fixed number of parameters**.
- Faster training.
- Assumes a specific data distribution.

Example:

- Linear Regression
- Logistic Regression

Non-Parametric Model

- Number of parameters is **not fixed**.
- More flexible.

- Can model complex relationships.

Example:

- KNN
- Decision Tree

Conclusion

Different Machine Learning models are selected based on the type of data and problem. Choosing the right model improves prediction accuracy and overall performance.

Q6) Explain Geometric Model and Probabilistic Model with Suitable Examples. (5 Marks)

1. Geometric Model

Definition

A Geometric Model represents data as **points in a geometric space** and separates different classes using **lines, planes, or decision boundaries**.

Working

- Every data item is represented as a point.
- The model measures the distance between points.
- A decision boundary separates different classes.

Applications

- Image classification
- Face recognition
- Pattern recognition

Example

Support Vector Machine (SVM)

It separates two classes using the **best possible boundary (hyperplane)**.

Class A ● ● ●

----- Decision Boundary

Class B ▲ ▲ ▲

Another example:

- K-Nearest Neighbors (KNN)

Advantages

- Easy to visualize.
- High accuracy for classification.
- Effective with numerical data.

Disadvantages

- Performance decreases with very high-dimensional data.
 - Sensitive to noisy data.
-

2. Probabilistic Model

Definition

A Probabilistic Model predicts the output by calculating the **probability** of different possible outcomes.

Working

- Calculates the probability for each class.
- Selects the class with the highest probability.

Applications

- Email spam detection
- Medical diagnosis
- Text classification

Example

Naive Bayes Classifier

Suppose an email has:

- Probability (Spam) = **0.90**
- Probability (Not Spam) = **0.10**

Since **0.90 > 0.10**, the email is classified as **Spam**.

Advantages

- Simple and fast.
- Works well with large datasets.
- Handles uncertain data effectively.

Disadvantages

- Assumes features are independent.
 - Accuracy depends on probability estimation.
-

Difference Between Geometric and Probabilistic Model

Geometric Model

Represents data as points in space

Uses distance and decision boundaries

Best for classification

Example: SVM, KNN

Probabilistic Model

Uses probability to predict outcomes

Uses probability values

Best for uncertain or noisy data

Example: Naive Bayes

Q1) Explain Geometric Models and Their Types. (Sept 2024 – 5 Marks)

Definition

A **Geometric Model** represents data as **points in geometric space**. It classifies data by drawing **lines, planes, or decision boundaries** between different classes.

Working

1. Convert data into points in space.
2. Measure the distance or similarity between points.
3. Draw a decision boundary to separate different classes.
4. Predict the class of new data.

Types of Geometric Models

1. Linear Geometric Model

- Uses a **straight line or hyperplane** to separate data.
- Suitable when data is linearly separable.

Example:

- Support Vector Machine (SVM)
 - Linear Regression
-

2. Distance-Based Geometric Model

- Classifies data based on the **distance** between data points.
- The nearest data points determine the class.

Example:

- K-Nearest Neighbors (KNN)
-

Applications

- Face recognition
- Image classification
- Pattern recognition
- Handwriting recognition

Advantages

- Easy to visualize.
- High accuracy for classification.
- Works well with numerical data.

Disadvantages

- Less effective with very high-dimensional data.
 - Sensitive to noisy data.
-

Q2) Describe Parametric and Non-Parametric ML Models. (Oct 2022 – 5 Marks)

Definition

Machine Learning models are classified into **Parametric** and **Non-Parametric** models based on the number of parameters they use.

Parametric Model

Definition

A **Parametric Model** has a **fixed number of parameters**. It assumes that the data follows a specific pattern or distribution.

Characteristics

- Fixed number of parameters.
- Faster training.
- Requires less data.
- Simpler model.

Examples

- Linear Regression
- Logistic Regression
- Naive Bayes

Advantages

- Fast training.
- Easy to implement.
- Less memory required.

Disadvantages

- Less flexible.
 - Performance decreases if assumptions are incorrect.
-

Non-Parametric Model

Definition

A **Non-Parametric Model** does **not have a fixed number of parameters**. It can learn more complex relationships from the data.

Characteristics

- Flexible model.
- Number of parameters depends on data.
- No assumption about data distribution.
- Requires more data.

Examples

- K-Nearest Neighbors (KNN)
- Decision Tree
- Random Forest

Advantages

- Handles complex data.
- Better accuracy for complex problems.
- Fewer assumptions.

Disadvantages

- Slower training.
- Requires more memory and computation.

Difference Between Parametric and Non-Parametric Models

Parametric Model

Fixed number of parameters

Faster training

Assumes data distribution

Less flexible

Needs less data

Examples: Linear Regression, Naive Bayes

Non-Parametric Model

Number of parameters is not fixed

Slower training

No assumption about data distribution

More flexible

Needs more data

Examples: KNN, Decision Tree

Q3) Describe Grouping and Grading Models. (Sept 2024 – 5 Marks)

Definition

Grouping and Grading models are Machine Learning models used to **organize** or **evaluate** data.

1. Grouping Model

Definition

A **Grouping Model** divides similar data into different groups or clusters based on their characteristics.

Characteristics

- Finds hidden patterns.
- No predefined labels are required.
- Mainly used in Unsupervised Learning.

Applications

- Customer segmentation
- Market basket analysis
- Document clustering

Example

- **K-Means Clustering**

2. Grading Model

Definition

A **Grading Model** assigns a **score, rank, or grade** to data based on its characteristics.

Characteristics

- Used for prediction and ranking.

- Helps in decision-making.

Applications

- Student grade prediction
- Credit score prediction
- Employee performance evaluation

Example

- Credit scoring system
- Student performance prediction

Difference Between Grouping and Grading Models

Grouping Model

Groups similar data

Used for clustering

Finds hidden patterns

Example: K-Means

Grading Model

Assigns score or rank

Used for prediction and evaluation

Measures performance

Example: Credit Score Prediction

Q) Write a Short Note on Learnability and Statistical Learning Approach. (5 Marks)

A) Learnability

Definition

Learnability is the ability of a Machine Learning algorithm to learn patterns from the training data and make accurate predictions for new or unseen data.

Factors Affecting Learnability

1. Quality of training data.
2. Quantity of training data.
3. Selection of the learning algorithm.
4. Number of features (attributes).
5. Presence of noise or missing values.

Importance

- Improves prediction accuracy.
- Helps the model generalize to new data.
- Reduces prediction errors.
- Produces reliable ML models.

B) Statistical Learning Approach

Definition

The **Statistical Learning Approach** is a Machine Learning approach that uses **statistics and probability** to learn patterns from data and make predictions.

Working

1. Collect data.
2. Analyze data using statistical methods.
3. Train the model.
4. Predict results for new data.

Applications

- Weather forecasting
- Stock market prediction
- Disease prediction
- Sales forecasting

Advantages

- Provides accurate predictions.
- Handles uncertain and noisy data.
- Suitable for large datasets.
- Supports better decision-making.

Difference Between Learnability and Statistical Learning

Learnability	Statistical Learning
Measures the ability of a model to learn from data.	Uses statistical methods to learn patterns and make predictions.
Focuses on the learning capability of the algorithm.	Focuses on data analysis and prediction.
Improves model performance and generalization.	Builds predictive models using statistical techniques.

Conclusion

Learnability determines how effectively an ML model learns from data, while the Statistical Learning Approach applies statistical techniques to build models that can accurately predict outcomes from data. Both are essential for developing efficient Machine Learning systems.

Q) Explain the Data Formats Used in Machine Learning (Multidimensional Data / Data Formats Conforming to ML Elements). (5 Marks)

Definition

Data Format is the way data is organized and represented before it is used to train a Machine Learning model. Proper data formatting helps the model learn efficiently and improve prediction accuracy.

Types of Data Formats

1. Structured Data

Definition:

Structured data is organized in a fixed format of **rows and columns**.

Examples:

- Excel sheets
- SQL databases
- Student records

Characteristics:

- Easy to store and search.
- Well organized.
- Suitable for most ML algorithms.

2. Semi-Structured Data

Definition:

Semi-structured data does not follow a fixed table format but contains **tags or key-value pairs** to organize information.

Examples:

- JSON
- XML
- HTML

Characteristics:

- Flexible structure.
- Easy to exchange between applications.
- Partially organized.

3. Unstructured Data

Definition:

Unstructured data has **no predefined format** and cannot be stored directly in tables.

Examples:

- Images
- Videos
- Audio files
- Emails
- Text documents
- Social media posts

Characteristics:

- Large amount of real-world data.

- Requires preprocessing before training.

Multidimensional Data

Definition

Multidimensional data contains **multiple features (attributes)** for each data record.

Example

Student Age Marks Attendance

A	20	85	90%
B	21	78	88%
C	20	92	95%

Here:

- **Age**
- **Marks**
- **Attendance**

are different **dimensions (features)** used by the ML model.

Importance of Data Formats in ML

- Improves model accuracy.
- Makes data preprocessing easier.
- Reduces training time.
- Helps in better prediction and decision-making.
- Supports efficient storage and processing of data.

Conclusion

Machine Learning models require data in a proper format. **Structured, Semi-Structured, Unstructured, and Multidimensional data** are commonly used depending on the application. Choosing the correct data format improves the performance and accuracy of the ML model.

Q) Describe Various Statistical Learning Approaches. (8 Marks / 5 Marks)

Definition

Statistical Learning is a Machine Learning approach that uses **statistics and probability** to learn patterns from data and make predictions or decisions. It helps in analyzing data, building models, and predicting future outcomes.

Types of Statistical Learning Approaches

1. Supervised Statistical Learning

Definition

Uses **labeled data**, where both input and correct output are available.

Purpose

- Predict future values.
- Perform classification and regression.

Examples

- House price prediction
 - Email spam detection
-

2. Unsupervised Statistical Learning

Definition

Uses **unlabeled data** to discover hidden patterns or groups.

Purpose

- Clustering
- Pattern discovery

Examples

- Customer segmentation
 - Market basket analysis
-

3. Parametric Statistical Learning

Definition

Uses models with a **fixed number of parameters** and assumes a specific relationship between variables.

Characteristics

- Simple and fast.
- Requires less training data.
- Easy to interpret.

Examples

- Linear Regression
 - Logistic Regression
-

4. Non-Parametric Statistical Learning

Definition

Uses models with **no fixed number of parameters** and does not assume any specific data distribution.

Characteristics

- Flexible.
- Handles complex data.
- Requires more training data.

Examples

- K-Nearest Neighbors (KNN)
 - Decision Tree
-

Working of Statistical Learning

1. Collect data.
 2. Clean and preprocess the data.
 3. Select an appropriate statistical model.
 4. Train the model using data.
 5. Test and evaluate the model.
 6. Use the model for prediction.
-

Applications

- Weather forecasting
 - Disease diagnosis
 - Stock market prediction
 - Sales forecasting
 - Credit risk analysis
-

Advantages

- Improves prediction accuracy.
 - Handles uncertainty in data.
 - Works well with large datasets.
 - Supports better decision-making.
-

Disadvantages

- Requires good-quality data.
 - Performance depends on the selected model.
 - Complex models need more computation.
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Conclusion

Statistical Learning approaches help Machine Learning systems understand patterns in data and make accurate predictions. Depending on the type of data and problem, **Supervised, Unsupervised, Parametric, and Non-Parametric** approaches are used.